

Molecular Content of Star-Forming Regions: Isotopic Ratios

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1 Introduction

During this internship, we focus on pre-stellar cores, at the very beginning of star formation, even before the star has formed. This implies very cold physical conditions (5 – 12 Kelvin) and low-density environments ($10^4 - 10^6 \text{ cm}^{-3}$). These sources are composed of gas (mainly H_2) and grains (Silicate). Measurements of isotopic ratios allow us to trace the elements and study the molecules present at the beginning of star formation, in order to understand the incorporation and destruction of molecules in subsequent phases of star formation. It is possible to study isotopic ratios using the ^{13}C isotope. These ratios could be a key to revealing the mechanisms of star formation.

We studied the pre-stellar core L1689b in the Ophiuchus molecular cloud using rotational transition observations measured by the 30-meter IRAM telescope.

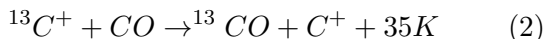
2 Challenges

The standard value of the isotopic ratio $[^{12}\text{C}]/[^{13}\text{C}]$ was determined by Lucas & Lizst (1998)^[1]:

$$\frac{[^{12}\text{C}]}{[^{13}\text{C}]} = 60 - 70 \quad (1)$$

The CO molecule in molecular clouds is the main reservoir of carbon. Therefore, when discussing the standard values of the $[^{12}\text{C}]/[^{13}\text{C}]$ isotopic ratio in the interstellar medium, astrophysicists use the ratio of CO .

Furthermore, chemical fractionation reactions can occur in the gas, particularly the exothermic reaction of ^{13}C depletion (2), which predominantly occurs in one direction due to the very low temperature of the pre-stellar core. This reaction depletes the gas in $^{13}\text{C}^+$ and traps it in CO .



We expect to obtain an isotopic ratio $[^{12}\text{C}]/[^{13}\text{C}]$ higher than the standard ratio for carbon chains

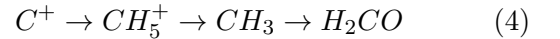
formed from C^+ , as detected by Sakai et al. (2010)^[2] on CCH and Yoshida et al. (2015)^[3] on C_3H_2 . We are interested in the abundance anomaly of carbon 13 in carbon chains.

Watanabe & Kouchi (2002)^[4] determined that CO , after forming in the gas phase, sticks to grains. We will therefore determine whether methanol (CH_3OH) is primarily formed from CO through successive hydrogenations on the grain by determining its isotopic ratio. We expect to obtain the same ratio as the standard value (1).

The hydrogenation pathway involves formaldehyde (H_2CO) on the grain, as shown in the reaction (3).



Additionally, a second formation pathway in the gas is also possible (4).



By studying the isotopic ratio of formaldehyde, we can determine the main formation pathway. If the primary pathway is the hydrogenation of CO on the grain, then we expect to obtain the standard ratio (1). On the other hand, if it is mainly formed in the gas (4), we expect to observe a depletion in ^{13}C , leading to a higher isotopic ratio.

3 Results

3.1 Carbon Chains

We studied two carbon chains: CCH and C_3H_2 , for which we determined their isotopic ratio $[^{12}\text{C}]/[^{13}\text{C}]$. For this study, we used a radiative transfer code (Radex). We employed a χ^2 method to compare the fluxes observed by the telescope with the fluxes calculated by the Radex model. The results are presented below (5).

$$\begin{aligned} \frac{[\text{CCH}]}{[\text{C}^{13}\text{CH}]} &= 125 \pm 13 & \frac{[\text{CCH}]}{[\text{C}^{13}\text{CH}]} &= 195 \pm 19 \\ \frac{[\text{C}^{13}\text{CH}]}{[\text{C}^{13}\text{CH}]} &= 1.6 \pm 0.2 \end{aligned} \quad (5)$$

We indeed observe the expected depletion of ^{13}C as predicted by (2). Additionally, we find that the

ratio between $C^{13}CH$ and ^{13}CCH is not equal to 1. This ratio could be explained by the following reaction (6), which is slightly more favorable in the right direction due to the very cold temperature conditions of the pre-stellar core.



We apply the same method to C_3H_2 , and we obtain (7).

$$\frac{[C_3H_2]}{[^{13}C_3H_2]} = 116 \pm 18 \quad (7)$$

We also obtain a characteristic ratio for carbon chains that is higher than the standard value (1).

3.2 CO and CH₃OH

To determine the isotopic ratio $[^{12}C]/[^{13}C]$ of CO , we calculate the line ratios. Since CO and ^{13}CO are optically thick, it is impossible to obtain their isotopic ratio directly. On the other hand, the molecules $C^{18}O$ and $^{13}C^{18}O$ are optically thin, so their isotopic ratio can be calculated using the line ratio between the rotational transition of $C^{18}O$ and that of $^{13}C^{18}O$. It is then possible to calculate the isotopic ratio $[^{12}C]/[^{13}C]$ as follows:

$$\frac{^{12}C}{^{13}C} = \frac{[C^{18}O]}{[^{13}C^{18}O]} \quad (8)$$

Thus, we obtain:

$$\frac{[C^{18}O]}{[^{13}C^{18}O]} = 52 \pm 2 \quad (9)$$

It is not possible to study the isotopic ratio of CO in the gas phase, as the molecules are already on the grains.

However, methanol is still present in the gas phase, so we can measure its isotopic ratio. We obtained the following ratio for methanol:

$$\frac{[^{12}C]}{[^{13}C]} = 73 \pm 13 \quad (10)$$

Thus, we obtain the standard isotopic ratio for methanol, corresponding to the one determined by Lucas & Lizst (1998)^[1]. Methanol is mainly formed from the hydrogenation of CO on the grain. Therefore, methanol that we observe in the gas phase has been desorbed and expelled by shocks (caused by the collapse of the pre-stellar core) from the grains.

The isotopic ratios obtained are summarized in Table 1 (1).

Molecule	Species	Isotopic Ratio $^{12}C/^{13}C$
CO	$^{12}C/^{13}C$	52 ± 2
CH_3OH	$^{12}C/^{13}C$	73 ± 13
CCH	$^{12}C/^{13}C$	125 ± 13
C_3H_2	$^{12}C/^{13}C$	116 ± 18

Table 1: Summary of Isotopic Ratios Obtained for L1689b.

4 Conclusion

The isotopic ratios determined in this work highlight the different pathways of molecular formation in the pre-stellar core L1689b in the Ophiuchus cloud. The results obtained for carbon chains (CCH , C_3H_2) confirm the depletion of ^{13}C in C^+ due to the fractionation reaction in CO (reaction (2)). Additionally, we obtain an isotopic ratio of CH_3OH consistent with the standard ratio, confirming that this molecule is mainly formed on grains.

The results are consistent with previous studies conducted on carbon chains by Sakai et al. (2010)^[2] and Yoshida et al. (2015)^[3], and on methanol by Watanabe & Kouchi (2002)^[4]. The isotopic ratios obtained provide valuable information about the initial conditions and chemical processes during star formation, offering insights into the origins of molecular complexity in the universe.

5 References

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